COVID-19 Detection from Chest X-ray Images Using Deep Learning Approach

***Abstract*—With the recent COVID-19 outbreak, rapid diagnostic testing has become a key challenge due to high rate of false positive in COVID-19 testing. The automatic detection and classification of Corona virus from the chest X-ray using computer aided technology can be a very useful addition to the less sensitive traditional method of detecting COVID-19. To enhance the performance of automated COVID-19 detection model, a CNN based model is proposed for the classification of COVID-19, normal and Pneumonia from X-ray images. For detecting COVID-19, biomedical imaging exams such as CT scans or CXR images can be used. As chest X-ray images are more widespread, faster and cheaper, datasets containing COVID positive, Normal and Pneumonia is being used in this work. Furthermore, to extract information from the lung region, lung segmentation method is implemented. Thus, this study aims to illustrate the effect of lung segmentation on the identification of COVID-19 based on CXR images. To segment the lung region from the plain CXR images, a U-Net CNN architecture has been proposed. For classification, several transfer learning Convolutional Neural Network (CNN) such as VGG19, DenseNet121, ResNet50, InceptionV2 and Xception were used. The models with their extensive filter family, abstraction and weight distribution capabilities can automatically reveal the discriminating features of the X-ray images. The performance was evaluated on both plain X-ray and lung segmented dataset. Overall, the classification of COVID-19 from the lung segmented dataset achieved an accuracy of 86% for binary and 79% for ternary classification. The proposed system is very reliable and has a comparable performance, improving the rapid and robust detection of COVID-19 using chest radiography.**

I. Introduction

The rise of severe acute respiratory syndrome coronavirus in Wuhan, China at the end of 2019 has created a large scale worldwide outbreak and is a severe public health issue [1]. COVID-19 is a novel viral infection that enters through the human's respiratory tract and affects the lungs. Various symptoms ranging from fever, dry cough, aches, diarrhea, shortness of breathing, organ failure has been detected from COVID-19 affected patients. On the scale of the global pandemic, the number of positive cases and deaths are increasing day by day [2].

With this recent COVID-19 outbreak, rapid diagnostic tests have become one of the main difficulties due to the severe scarcity of test kits [3]. Because of the limited accessibility of COVID-19 test kits, we search for other symptomatic measures. Meanwhile, COVID-19 assaults the epithelial cells that line our respiratory lot. And in such cases X-rays can be used to analyze the lungs of a patient [4]. RT-PCR test is very popular but the rate of false negative is relatively high. An alternate tool for testing is CT scan. Although CT scans can provide finer results, it is costly, consumes much time and contains high radiation. Compared with computed tomography (CT), diagnostic ultrasound, and magnetic resonance imaging (MRI), X-ray is easier to access, less expensive, faster and has low radiation than CT scanning. Chest X-beam is the most well-known test among the clinical imaging modalities. Much research has been made on chest X-ray classification for detecting various lung diseases. It is particularly applied to the detection and differentiation of lung cancer, tuberculosis, pneumonia and COVID-19 as well.

A. M. Ismael [2], proposed three deep CNN approaches for automatic detection of COVID-19 from X-ray images. Deep feature extraction using SVM classifier, fine tuning and for end to end training new CNN model has been experimented. ResNet18, ResNet50, ResNet101, VGG16, and VGG19 pertained CNN models were used. Moreover, for deep features classification support vector machine classifier was implemented with various kernel function like linear, quadratic, gaussian etc.

A deep learning model has been approached to predict the structural abnormalities from the chest X-ray images in [7]. Also, transfer learning was proposed to predict the disease from posterior and anterior radiographs. The proposed method achieved an accuracy of 96.3%. However, no preprocessing techniques were applied to attain a better performance in training the model. Again, the authors in [8] developed a deep learning framework including seven architectures for prediction of COVID-19 and healthy X-rays. The authors reported a comparative performance of the deep learning model to show a COVID-19 classification accuracy using a small number of datasets. The authors claimed that the VGG19 and DenseNet201 showed the highest accuracy 90%. Another study [9], used X-ray images of COVID-19 and healthy patients to detect COVID positive using existing CNN models like MobileNetV2, AlexNet, DenseNet, VGG16, VGG19, InceptionV3, NFNet, ResNet50, ResNet101, EfficientNetB7, and GoogLeNet. Among all the CNN models, MobileNetV2 achieved the highest accuracy for COVID classification which followed by VGG19 and ResNet101. Although one of their limitations is that no multi class classification model is used to detect pneumonia.

Using a very small amount of COVID-19 dataset, proposed a deep convolutional neural network based architecture called CovXNets to detect COVID-19 and other pneumonia diseases. For extracting diversified features, depth wise convolution with varying dilation rates has been utilized. As COVID-19 and pneumonia have many overlapping features, it acquired a smaller accuracy for detecting COVID-19 and Viral Pneumonia [3]. However, in their proposed method, different

variety of CovXNets are developed and trained with X-ray dataset of several resolutions to optimize their predictions. Also, stacking algorithm is being implemented in the model. In the end, a gradient based discriminative localization is employed to distinguish the abnormal regions of the chest X-ray images. However, the authors in [6], showed some concerns regarding practical application of the characterization of X-ray images utilizing various models. They tried to train several datasets and showed that the network performance dropped when the model was trained on one dataset and tested on another. In [10], a deep transfer learning based pre-trained model was approached to automatically detect COVID-19 and other pneumonia patients where ResNet50 got the highest accuracy which is 96.1%. Besides, three different binary classification with four classes (Covid-19, Normal, viral pneumonia and bacterial pneumonia) has been implemented using five-fold cross validation. Moreover, transfer learning techniques were applied using ImageNet data to reduce the training time. Another study came forward with SqueezeNet network that is tuned with Bayesian optimization additive [11]. The network performance is improved using fine tuned hyper-parameters and augmented dataset. A multi scale augmentation is performed to overcome the limitations of the uneven distribution of the dataset. A CNN based classifier based on Split-Transform-Merge (STM) is proposed that is capable to capture diversified features at various levels for screening COVID-19 from X-ray [12]. The authors in [13] proposed deep CNN network using a public dataset. For improving the performance the model is developed with residual learning and squeeze excitation.

The majority of the research papers that are dealing with COVID-19 classification, report extremely superior results in this work. However, Cohen et al. reported the limitations of the X-ray image classification because of the fact that the model can learn features specific to the data set instead of the features specific to the lung infection [5] [6]. In order to increase the efficiency or accuracy for the detection of a lung disease, identifying the lung nodule is an essential step. This has been the primary motivation for conducting the work. The key contribution of this work is to deal with lung segmentation to identify the lung region. And implementing the CNN models with the segmented lung region. The primary purpose of this work is to develop an efficient Computer Aided Diagnosis System for classification of COVID-19 from chest X-ray images.

II. Materials and methods

A. Dataset Description

1) Segmentation dataset: For automatic segmentation, a CNN based architecture called U-Net has been implemented. To train the U-Net model, we used Montgomery County datasets that consists of gold standard binary mask. This X-ray set consists of 138 images including 80 normal and rest of them abnormal with infection of tuberculosis. Each of the X-ray consists of ground truth which are manually generated.

2) Classification dataset: For classification phase, two dataset has been utilized for evaluating the performance on both binary and three class classification.

• First dataset named as COVIDGR consists of total 852 chest X-ray images. Out of them 426 are COVID positive and rest 426 are Non-COVID X-rays. All the X-ray images are posterior anterior (PA) view and were taken from the same X-ray machine. All the positive images were labelled as COVID-19 only when tested positive by RT-PCR and confirmed by expert radiologists. Moreover, the positive images comprises of several severity level from mild to severe cases.

• The second dataset is collected from the kaggle source and is used for evaluating the performance for three class classification specifically (COVID-19, Normal & Pneumonia). Each class consists of 821 X-ray images.

B. Proposed System

To understand the proposed system, Fig. 1 represents the general overview of the proposed system. The approached system comprises of two parts- 1. Lung segmentation 2. Classification (Segmented and plain X-ray).

The experiment is carried out by using U-Net model for separating the lung region from the whole X-ray image and creating a new dataset of the lung segmented region. And evaluating the both the segmented and original X-ray dataset for classification of several deep neural network approaches- VGG19, DenseNet121, ResNet50, InceptionV2 and Xception.

1) Lung Segmentation Phase: For biomedical image segmentation, the U-Net architecture is proposed in [14], which showed much popularity due to its high accuracy in segmentation process. Ahmed, S., et al. in [15] made a comparative research and showed that among all the segmentation models (DenseNet, SegNet, DeepLab, VB-Net, NABLA-N, U-Net), U-Net has been resulted in a significant adoption as a prime tool for lung segmentation of CXR or CT scan images in majority of the cases. Gite et al. [16] made a comparison of U-Net and its variants with various segmentation models like- CN and SegNet. And showed that U-Net got the highest accuracy for segmenting X-ray image for pulmonary lung diseases.

The plain X-ray dataset of COVID-19, Pneumonia and Normal X-ray does not contain manually segmented lung mask. Hence, we approached an automated segmentation process for creating lung mask for every X-ray images of the plain X-ray dataset.

When a input is given to the trained U-Net model, the model predicts the binary mask of the given input. After a post processing step, the output becomes more precise and gives a clear edge of the binary mask. Then bitwise_and operation is performed between the original and predicted binary mask. After the bitwise_and operation the final lung mask is obtained.

After creating the U-Net model from scratch, to improve the model performance, several hyperparameter tuning has been experimented on batch size, optimizer and learning rate. We got the best output by using batch size 16, optimizer 'Adam' and learning rate 0.0005.

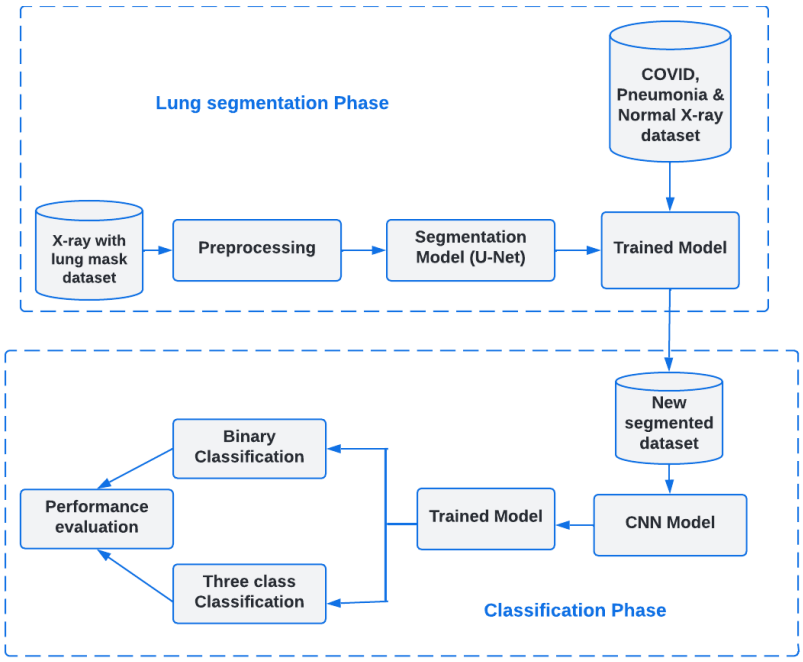

Fig. 1. Overview of the system

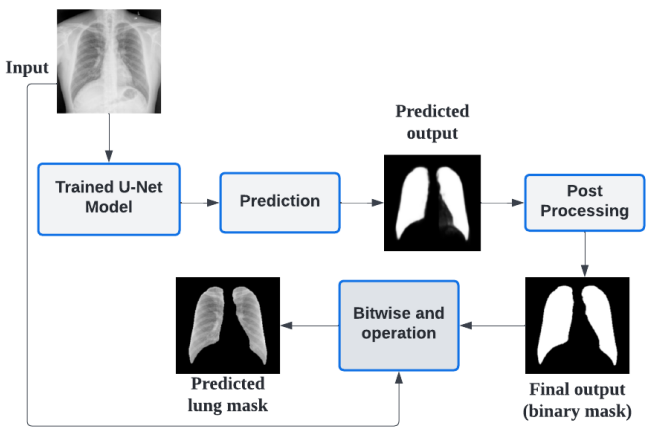

Fig. 2. Overview of segmentation process

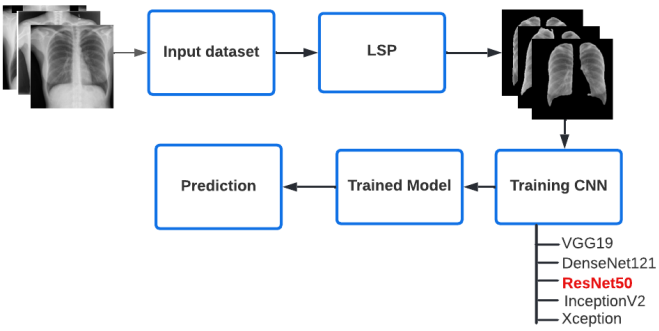

Fig. 3. Overview of classification process

2) Classification Phase: Classification is the process of extracting features from the dataset images to recognize some patterns in them. Deep learning also called feed forward network, process output by back propagation. This a supervised learning based classification network where the data flow is unidirectional.

Fig. 3 shows the overall overview of the classification phase for COVID-19 detection. After the lung segmentation phase (LSP), the model gives the segmented dataset. The segmented dataset is used for training the CNN model. For evaluating the performance for segmented and non-segmented dataset, the plain X-ray images is also trained to with same CNN model. Several transfer learning approaches has been applied to get the highest accuracy.

Activation function in CNN is used to learn complex pattern within the network and delivers weighted value of the input to the output through nodes. In simple words, it helps to give an output depending on the input in the network. In this network for hidden layers relu is used and for output layer sigmoid is used as the activation function. The function optimizer in neural network is used to modify the weights and learning rate of the model. Adam optimized is being implemented with learning rate of 5*1e-4. A dropout layer with probability of 0.5 is used in the model to prevent overfitting. The function of dropout function is to select some neurons randomly at training time to reduce the loss rate. A batch size of 16 is used in the network during training.

While training the model, we used three keras call back function to interact with the training process for achieving a better performance.

1. Early stopping: Early stopping is a process of choosing randomly large number of epoch during the training period. It basically monitors the validation accuracy or loss while training. When the network performance stops improving or remains constant for the number off patience given, the training automatically halts. This helps to reduce the task of taking different number of epoch and fine tune the model over and over again. For this model, the patience value has been set to 10. So, if the validation accuracy stops improving after consecutive 10 epochs then training stops.

2. Learning rate scheduler: This keras function is used to adjust or reduce the learning rate of the model overtime. This helps the model to learn good weights earlier. The function monitors the validation accuracy up to a certain patience and if no improvement occurs, it reduces the learning rate to a certain amount. In this model, a patience value of 10 and the minimum learning rate is used 0.0001.

3. Model checkpoint: In case of any interruption the function is used to resume the training process. And the model weights are saved to a checkpoint file at certain intervals. So the model is saved and can be loaded any time later when needed. In this network, the save_best_only=True is set. So, the function monitors the validation accuracy and save the best weights only during the training process.

For classification phase, two dataset has been utilized for evaluating the performance on both binary and three class classification. We used five different transfer learning model and ResNet50 achieved the highest accuracy for both segmented and non-segmented dataset.

III. Results and Discussions

In this section, we will evaluate the performance of the proposed model with different datasets. And consists of the overall descriptions of the implementation result analyses of the proposed framework.

A. Lung Segmentation

The trained U-Net model from scratch is used to segment the two classification dataset which is being used for binary and three class classification of COVID-19. Each images of the datasets are passed to the trained U-Net model to produce binary mask of the input images. We are able to achieve an average 90% dice coefficient for generating lung mask from X-rays. Bitwise_and operation is performed to create the the lung contour of the original images and thus new dataset is being created for the performing classification between COVID, Normal Pneumonia. It's very important to see how the trained U-Net model works on unseen data with COVID-19 lung infection. It can be seen from the Fig. 4 that, the segmentation model works very well for identifying the lung region for COVID, Normal and Pneumonia of the classification dataset. A qualitative evaluation was done to confirm that each of the images of the dataset was segmented accurately.

B. COVID-19 classification

As mentioned above, two types of experiment were performed (segmented & plain dataset) for both binary and three class classification. The model using segmented dataset achieved the highest accuracy than the plain X-ray dataset. We used five different transfer learning model and ResNet50 achieved the highest accuracy for segmented dataset. In the following, a detailed performance evaluation for both binary and three class classification is represented below:

TABLE I
Performance comparison of various models on binary classification (COVID & Non-COVID)

Pretrained Models	Dataset	Accuracy	Precision	Recall	F1
VGG19	Segmented	0.83	0.84	0.83	0.84
	Non-segmented	0.79	0.81	0.80	0.80
DenseNet121	Segmented	0.80	0.83	0.79	0.81
	Non-segmented	0.75	0.74	0.75	0.75
ResNet50	**Segmented**	**0.86**	**0.87**	**0.83**	**0.85**
	Non-segmented	0.83	0.85	0.82	0.83
InceptionV2	Segmented	0.81	0.77	0.75	0.76
	Non-segmented	0.76	0.73	0.75	0.74
Xception	Segmented	0.75	0.75	0.74	0.75
	Non-segmented	0.73	0.77	0.70	0.73

TABLE II
Performance comparison of various models on three class classification (COVID, Normal & Pneumonia)

Pretrained Models	Dataset	Accuracy	Precision	Recall	F1
VGG19	Segmented	0.77	0.75	0.73	0.74
	Non-segmented	0.73	0.73	0.70	0.71
DenseNet121	Segmented	0.77	0.74	0.73	0.74
	Non-segmented	0.70	0.68	0.71	0.69
ResNet50	**Segmented**	**0.79**	**0.79**	**0.78**	**0.78**
	Non-segmented	0.75	0.76	0.72	0.74
InceptionV2	Segmented	0.75	0.78	0.75	0.76
	Non-segmented	0.71	0.68	0.67	0.67
Xception	Segmented	0.72	0.75	0.69	0.72
	Non-segmented	0.69	0.64	0.68	0.66

The five pretrained model performance on both binary and multicalss classification is represented in table I and table II. We see that for both the classification ResNet50 achieved the maximum accuracy.

When a model is trained with full X-ray images, it learns background noise and tends to make the model overfit. Also, false prediction might increase as the model gets background information other than lung. In images segmentation, separating the lung nodule from the whole X-ray image, forces the model to get information from the lung region only. As a result, the tendency of false prediction or overfitting might

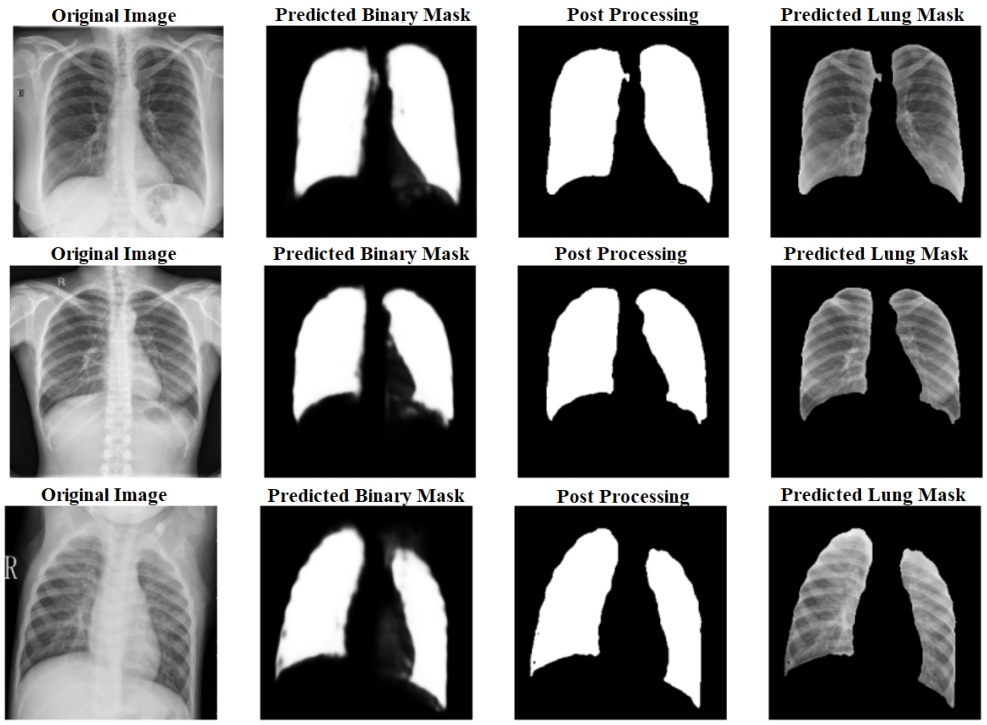

Fig. 4. Lung mask of the classification dataset: COVID-19(top row), Normal(middle row), Pneumonia(bottom row)

reduce to an extent. To access the impact of lung segmentation, to detect COVID-19 and pneumonia disease, we compared the result for both segmented and non segmented dataset. We got a better accuracy from the segmented dataset compared to non-segmented (plain X-ray) dataset. More correct predictions comes from the segmented dataset. From the table III and table IV, we see that for both binary and multiclass classification segmented dataset got the highest accuracy compared to plain X-ray(non-segmented) dataset.

TABLE III
COMPARATIVE CLASSIFICATION PERFORMANCE OF SEGMENTED & NON-SEGMENTED DATASET FOR TWO CLASS

Dataset (COVID & Normal)	Accuracy	Precision	Recall	F1
With Segmentation	**0.86**	**0.87**	**0.83**	**0.85**
Without Segmentation	0.83	0.85	0.82	0.83

TABLE IV
COMPARATIVE CLASSIFICATION PERFORMANCE OF SEGMENTED & NON-SEGMENTED DATASET FOR THREE CLASS

Dataset (COVID, Normal & Pneumonia)	Accuracy	Precision	Recall	F1
With Segmentation	**0.79**	**0.79**	**0.78**	**0.78**
Without Segmentation	0.75	0.76	0.72	0.74

Much research has been made on COVID-19 and other pneumonia diseases. But a small strides have been made on lung segmentation for effective classification of COVID-19. We can now compare the different works that involves COVID-19 detection. A comparative analysis of our proposed system with other studies is represented in this section. Table V shows the comparative analysis of the proposed system for binary classification with other works. The proposed model outperformed the other works using segmented dataset.

Our segmentation based CNN model brings a reliable result for classification of COVID-19 from X-ray images. Our COVID-19 detection model focuses on the lung region only for a reliable classification. Our lung segmentation approach reached a dice coefficient value of 90% which shows a robust segmentation output. Furthermore, in classification technique the model performed well using segmented dataset and achieved 86% for binary and 79% accuracy for multi class classification.

IV. CONCLUSION

In this paper, we proposed a computer aided COVID-19 classifier by extracting the lung region from the whole X-ray image. The performance of several CNN models were evaluated for both binary and multi class datasets. For segmented dataset, the model achieved more accuracy compared to plain X-ray dataset. Thus, the proposed system reaffirm the importance of the lung segmentation for reliable classification of COVID-19 and other pulmonary diseases.

As a future work, we aim to improve the image segmentation model that can able to detect the accurate lung boundary for the abnormal lung shapes and occluded X-ray images. And

TABLE V
COMPARATIVE ANALYSIS WITH OTHER WORKS

Related Papers	Dataset	Methods	Accuracy	F1
Tabik, Siham, et al. [17] 2020	COVIDGR	CNN based COVID-SDNet segmentation based cropping	76.18%	76.94%
Fang, Zhenyu, et al. [18] 2021	COVIDGR	CNN based MSRCovXNet	82.2%	83.46%
Alshehri, Elaf, et al. [19] 2022	COVIDGR	CNN	85%	84.7%
Zhao, Huimin, et al. [20] 2022	COVIDGR	CNN based SC2Net	84.23%	81.31%
Proposed Work	COVIDGR	CNN With segmentation	86%	85%

applying the segmentation process to detect other lung diseases like lung cancer, tuberculosis etc.

REFERENCES

[1] Lai, C. C., Shih, T. P., Ko, W. C., Tang, H. J., Hsueh, P. R. (2020). Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and coronavirus disease-2019 (COVID-19): The epidemic and the challenges. International journal of antimicrobial agents, 55(3), 105924.

[2] Ismael, A. M., Şengür, A. (2021). Deep learning approaches for COVID-19 detection based on chest X-ray images. Expert Systems with Applications, 164, 114054.

[3] Mahmud, T., Rahman, M. A., Fattah, S. A. (2020). CovXNet: A multi-dilation convolutional neural network for automatic COVID-19 and other pneumonia detection from chest X-ray images with transferable multi-receptive feature optimization. Computers in biology and medicine, 122, 103869.

[4] Salman, F. M., Abu-Naser, S. S., Alajrami, E., Abu-Nasser, B. S., Alashqar, B. A. (2020). Covid-19 detection using artificial intelligence.

[5] Maguolo, G., Nanni, L. (2021). A critic evaluation of methods for COVID-19 automatic detection from X-ray images. Information Fusion, 76, 1-7.

[6] Cohen, J. P., Hashir, M., Brooks, R., Bertrand, H. (2020, September). On the limits of cross-domain generalization in automated X-ray prediction. In Medical Imaging with Deep Learning (pp. 136-155). PMLR.

[7] Vaid, S., Kalantar, R., Bhandari, M. (2020). Deep learning COVID-19 detection bias: accuracy through artificial intelligence. International Orthopaedics, 44(8), 1539-1542.

[8] Hemdan, E. E. D., Shouman, M. A., Karar, M. E. (2020). Covidx-net: A framework of deep learning classifiers to diagnose covid-19 in x-ray images. arXiv preprint arXiv:2003.11055.

[9] Akter, S., Shamrat, F. J. M., Chakraborty, S., Karim, A., Azam, S. (2021). COVID-19 detection using deep learning algorithm on chest X-ray images. Biology, 10(11), 1174.

[10] Narin, A., Kaya, C., Pamuk, Z. (2021). Automatic detection of coronavirus disease (covid-19) using x-ray images and deep convolutional neural networks. Pattern Analysis and Applications, 24(3), 1207-1220.

[11] Ucar, F., Korkmaz, D. (2020). COVIDiagnosis-Net: Deep Bayes-SqueezeNet based diagnosis of the coronavirus disease 2019 (COVID-19) from X-ray images. Medical hypotheses, 140, 109761.

[12] Khan, S. H., Sohail, A., Khan, A., Lee, Y. S. (2022). COVID-19 detection in chest X-ray images using a new channel boosted CNN. Diagnostics, 12(2), 267.

[13] Agrawal, T., Choudhary, P. (2022). FocusCovid: automated COVID-19 detection using deep learning with chest X-ray images. Evolving Systems, 13(4), 519-533.

[14] Ronneberger, O., Fischer, P., Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer-assisted intervention (pp. 234-241). Springer, Cham.

[15] Shah, F. M., Joy, S. K. S., Ahmed, F., Hossain, T., Humaira, M., Ami, A. S., ... Ahmed, S. (2021). A comprehensive survey of COVID-19 detection using medical images. SN Computer Science, 2(6), 1-22.

[16] Gite, S., Mishra, A., Kotecha, K. (2022). Enhanced lung image segmentation using deep learning. Neural Computing and Applications, 1-15.

[17] Tabik, S., Gómez-Ríos, A., Martín-Rodríguez, J. L., Sevillano-García, I., Rey-Area, M., Charte, D., ... Herrera, F. (2020). COVIDGR dataset and COVID-SDNet methodology for predicting COVID-19 based on chest X-ray images. IEEE journal of biomedical and health informatics, 24(12), 3595-3605.

[18] Fang, Z., Ren, J., MacLellan, C., Li, H., Zhao, H., Hussain, A., Fortino, G. (2021). A novel multi-stage residual feature fusion network for detection of COVID-19 in chest X-ray images. IEEE Transactions on Molecular, Biological and Multi-Scale Communications, 8(1), 17-27.

[19] Alshehri, E., Kalkatawi, M., Abukhodair, F., Khashoggi, K., Alotaibi, R. (2022). COVID-19 Diagnosis from Medical Images Using Transfer Learning. Saudi Journal of Health Systems Research, 1-8.

[20] Zhao, H., Fang, Z., Ren, J., MacLellan, C., Xia, Y., Li, S., ... Ren, K. (2022). SC2Net: a novel segmentation-based classification network for detection of Covid-19 in chest X-ray images. IEEE Journal of Biomedical and Health Informatics, 26(8), 4032-4043.